

Project Report

Advanced Generative AI for Fashion Design

1. Executive Summary

The Generative Fashion Designer project is a production-grade, modular deep learning system designed to push the boundaries of textile and fashion synthesis. By implementing six state-of-the-art generative architectures, the system provides a comprehensive platform for researchers and designers to explore latent space representations of fabric textures and garments. The project integrates local GPU-accelerated training with cloud-based high-fidelity generation and a robust CI/CD deployment pipeline.

2. System Architecture

The system is built on a modular architecture to ensure scalability and ease of experimentation:

- Core Models: Custom implementations of VAE, DCGAN, WGAN-GP, and Diffusion Transformers.
- Data Pipeline: Sophisticated preprocessing, augmentation, and stratification for the DTD dataset.
- Infrastructure: Dockerized environment with automated CI/CD via GitHub Actions.
- Frontend: A modern, glassmorphic web UI for real-time interaction and cloud generation.

3. Deep Learning Models

3.1 beta-VAE (Variational Autoencoder)

Implements a Beta-VAE with KL-annealing. This model excels at learning disentangled latent representations of fashion textures. Key features include a residual encoder/decoder and reparameterization trick for stable backpropagation.

3.2 D

Adversarial models optimized for high-frequency details. DCGAN uses Spectral Normalization for discriminator stability, while WGAN-GP implements a Gradient Penalty to enforce the Lipschitz constraint, effectively eliminating mode collapse.

3.3 L

A modern approach using Transformers in a latent diffusion space. This model provides superior generative quality by leveraging the scalability of self-attention mechanisms.

4. Dataset: Describable Textures (DTD)

To simulate real-world textile design, the project utilizes the DTD dataset from Oxford VGG. It consists of 5,640 images across 47 categories, ranging from 'knitted' and 'braided' to 'paisley' and 'marbled'. This dataset serves as a proxy for high-resolution fabric textures.

5. Deployment and DevOps

The project is fully production-ready:

- Docker: The entire server is containerized for consistent deployment.
- CI/CD: GitHub Actions pipeline with Lint -> Test -> Deploy stages.
- LFS: Large model weights (up to 75MB) are managed via Git LFS.
- Hugging Face: Deployed as a Docker Space with secret-managed API keys.

6. Conclusion

The Generative Fashion Designer successfully demonstrates the integration of advanced generative AI with modern software engineering practices. It provides a robust foundation for future extensions in virtual try-ons and automated garment design.